

Sunburn device: the whole program

every block of the program that runs on the wrist, explained in plain English · printed 2026-09-14

The Sunburn device is a wearable built on a BBC micro:bit, a small computer with 25 lights and two buttons. It measures the sun's UV with a sensor, checks the online UV index for where you are through a phone, and then flashes, beeps or taps your arm until you put sunscreen on. Two hours later it reminds you to reapply. The program that does all this is built from 74 coloured blocks. Every one of them is on these sheets, with a note saying what it does.

Each picture is one block of the program, exactly as it looks in the MakeCode editor where the program is built by snapping coloured pieces together. A block is read from top to bottom. Under each picture: the block's name, what kind of block it is, what it does, which other blocks use it (Used by) and which blocks it calls on (Uses). This sheet is the whole program at a glance: nine numbered sections. Sections 1 and 2 are the plan the device follows; sections 3 to 9 are the parts that plan calls on.



- 1

Start here: settings · 1 block · sheet 2
This block runs first, the moment the device is switched on. It holds every setting: which UV sensor is fitted, how the sound and the tapping arm are wired, the UV levels that count as low, moderate, very high and extreme, and how long each wait is. Then it starts everything up. To change how the device behaves, this is the only place to look.
- 2

The flowchart · 20 blocks · sheet 2
The plan the device follows all day, taken straight from the Sunburn Flowchart. Every box and every question on the flowchart is one block here, with the same name, in the same order: read the UV sensor, check the reading makes sense, ask the phone for the online UV index, decide how strong the sun is, then react. The forever block keeps the plan running round and round.
- 3

Buttons · 3 blocks · sheet 2
The micro:bit has two buttons. A means done: sunscreen is on, or I am going inside. B wakes the device up. Pressing both together turns it off, or on again.
- 4

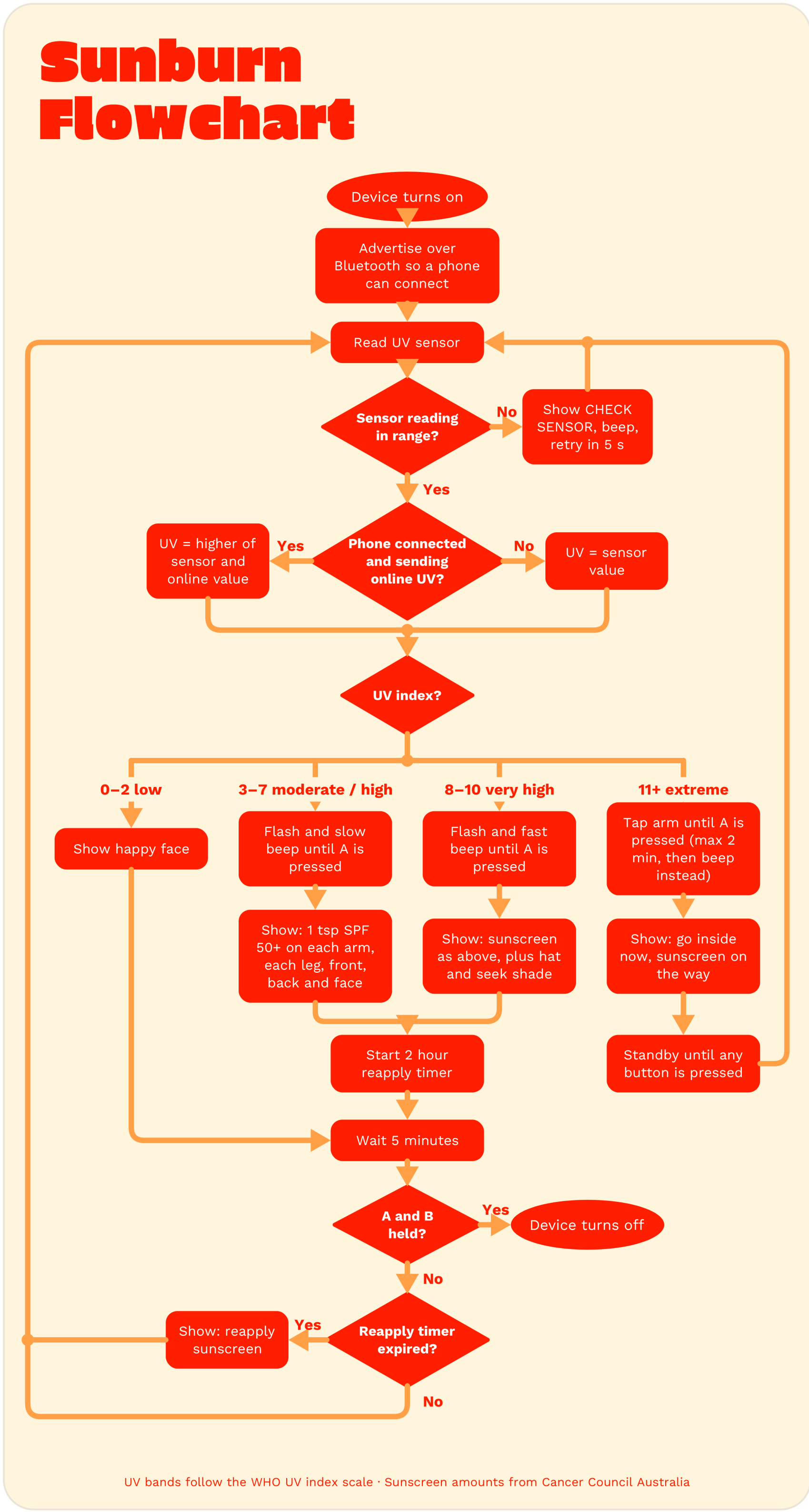
Alerts · 4 blocks · sheet 3
The ways the device gets your attention: flashing and beeping, tapping your arm with the small motor, and going to sleep when nobody answers. Each one keeps going until you press A or its time limit runs out. No answer for a long time means the device is probably not being worn, so it sleeps until a button is pressed.
- 5

Screen and lights · 8 blocks · sheet 3
What you see. The 5 by 5 grid of lights shows the UV number, a happy face or a warning, and messages one word at a time. If a small screen is fitted, it shows the number large, the level, the message and the countdowns.
- 6

Sound and the tapping arm · 5 blocks · sheet 3
The beeps, and the small motor (a servo) whose arm taps your wrist when the sun is extreme.
- 7

Talking to the phone · 11 blocks · sheet 3
How the wearable and the phone app talk over Bluetooth. The phone sends the online UV index for where you are; the wearable sends back what it is doing every two seconds and reports each event, such as sunscreen applied. The same messages also go down the USB cable, so everything can be tested on a laptop.
- 8

Helpers · 10 blocks · sheet 4
Small jobs the other blocks share: the maths for each type of sensor, turning minutes into the milliseconds the program counts in, sorting numbers, and tidying up numbers, times and text.



The Sunburn Flowchart the device follows. Every box on it is a block in section 2 with the same name.

What the block colours mean

- | | |
|-------------------------------|-----------------------------|
| Variables: remember a value | Text |
| Functions: a step with a name | Arrays: lists |
| Basic: the lights, pauses | Music: beeps |
| Input: buttons and time | Pins: sensor, motor, screen |
| Loops: repeat | Bluetooth: the phone |
| Logic: if, compare | Serial: the USB cable |
| Maths | Control: timing |

Reading the other sheets

Every picture is one block from the editor at 70% of its size on screen. Under it: the block's name, what kind of block it is, what it does, which blocks use it (Used by) and which blocks it calls on (Uses).

UV index How strong the sun's burning rays are, from 0 (none) to 11 and above (extreme). Protection is needed from 3 upwards, even on cool or cloudy days.

micro:bit The small computer the program runs on. It has a 5 by 5 grid of lights, buttons A and B, and sockets called pins (P0, P1, P2) that other parts plug into.

Block One instruction of the program. Blocks snap together and run in order from top to bottom. A block with a name at the top (function) is a step that other blocks can call on.

Sensor The part that measures UV. It gives a voltage, which the program turns into a UV index number.

Servo A small motor with an arm. Here it taps the wearer's wrist when the sun is extreme.

Bluetooth and USB The two ways the wearable talks to a phone or a laptop. The phone app sends the online UV index; the wearable sends back what it is doing.

OLED An optional small screen. Without one, the grid of lights does the job.

ms and Hz ms are milliseconds, thousandths of a second. Hz sets how high a beep sounds.

Demo timings Much shorter waits (seconds instead of minutes and hours) so the whole day can be shown in a few minutes at a demo.

Names in the blocks Words like uvSampleCount are the program's own names for its settings and memory. Read them as English: uv sample count is how many readings to take.

1

Start here: settings

This block runs first, the moment the device is switched on. It holds every setting: which UV sensor is fitted, how the sound and the tapping arm are wired, the UV levels that count as low, moderate, very high and extreme, and how long each wait is. Then it starts everything up. To change how the device behaves, this is the only place to look.

1 block

```
on start
  set uvSensorType to 1
  set uvSampleCount to 15
  set soundOutput to 1
  set servoType to 1
  set servoRestAngle to 90
  set servoTapAngle to 30
  set oledMode to 1
  set oledType to 1
  set oledAddress to 60
  set uvLowMax to 2
  set uvHighMax to 7
  set uvVeryHighMax to 10
  set waitMinutes to 5
  set reapplyHours to 2
  set tapMaxMinutes to 2
  set alertGiveUpMinutes to 10
  set onlineUvMaxAgeMinutes to 20
  set sensorRetrySeconds to 5
  set firmwareVersion to '3.1'

  call applySensorPreset
  call applyTimings
  call setupSound
  call setupOled
  call advertiseOverBluetooth

  if button B is pressed then
    set demoMode to true
    call applyTimings

  call deviceTurnsOn
```

on start Runs once, when the device is switched on. The first thing that runs when the device is switched on. All the settings are listed here, then the start-up steps: work out the sensor maths, set the timings, get the sound, the screen and Bluetooth ready, and show that the device is on. Holding B while switching on uses the fast demo timings.

Uses: deviceTurnsOn, setupSound, advertiseOverBluetooth, applySensorPreset, applyTimings, setupOled

2

The flowchart

The plan the device follows all day, taken straight from the Sunburn Flowchart. Every box and every question on the flowchart is one block here, with the same name, in the same order: read the UV sensor, check the reading makes sense, ask the phone for the online UV index, decide how strong the sun is, then react. The forever block keeps the plan running round and round.

20 blocks

```
forever
  if deviceOn then
    call runFlowchart
  else
    pause (ms) 500
```

forever Runs again and again, all day.

Runs the flowchart once, then goes straight back to the top and runs it again, all day long.

Uses: runFlowchart

```
function runFlowchart
  call readUvSensor
  if call sensorReadingInRange then
    call chooseUvValue
    call showUvReading
    set currentBand to call uvBand, currentUv
    set wentToStandby to false
    if currentBand == 'low' then
      call showHappyFace
    else if currentBand == 'modhigh' then
      call moderateOrHighUv
    else if currentBand == 'vhigh' then
      call veryHighUv
    else
      call extremelyUv
    +
    if deviceOn and not wentToStandby then
      call wait5Minutes
      if deviceOn and call reapplyTimerExpired then
        call showReapplySunscreen
      +
    else
      call showCheckSensor
```

runFlowchart A step of the program.

The whole flowchart from top to bottom. Read the sensor. If the reading makes sense, pick the UV value, show it, decide the level and react: a happy face, or an alert until A is pressed. Then wait 5 minutes and check whether it is time for more sunscreen. If the reading does not make sense, show CHECK SENSOR instead. Pressing A and B together at any moment turns the device off and stops every wait.

Used by: forever **Uses:** readUvSensor, sensorReadingInRange, showCheckSensor, chooseUvValue, uvBand, showHappyFace, moderateOrHighUv, veryHighUv, extremeUv, wait5Minutes, reapplyTimerExpired, showReapplySunscreen, showUvReading

```
function sunscreenApplied level
  show icon
  set protectionLevel to level
  change sunscreenCount by 1
  call sendEvent 'sunscreen'
  call start2HourReapplyTimer
```

sunscreenApplied A step of the program - needs: level.

A was pressed after a sunscreen alert: remembers which advice was given, counts one more application, tells the app, and starts the timer.

Used by: moderateOrHighUv, veryHighUv **Uses:** start2HourReapplyTimer, sendEvent

```
function alreadyProtected
  set stateName to 'protected'
  show icon
  set timeLeft to reapplyDueTime - running time (ms)
  call oledShowMessage join 'Sunscreen on. Reapply in' call formatTime timeLeft
```

alreadyProtected A step of the program.

Sunscreen is already on, so there is no alert: just a tick and the time until it needs reapplying.

Used by: moderateOrHighUv, veryHighUv **Uses:** oledShowMessage, formatTime

```
function readUvSensor
  set stateName to 'reading'
  repeat uvSampleCount times
    do
      samples = add value analog read pin analog pin P1 to end
      pause (ms) 2
  call sortNumbers samples
  set sensorRaw to samples get value at floor uvSampleCount / 2
  set sensorNoise to samples get value at uvSampleCount - 1 - samples get value at 0
  set sensorVolts to sensorRaw * 3.3 / 1023
  set sensorUv to sensorVolts - uvZeroVolts / uvVoltsPerIndex
  call debugLog join 'raw' sensorRaw 'noise' sensorNoise
```

readUvSensor A step of the program.

Reads the UV sensor plugged into pin P1. It takes 15 quick readings and uses the middle one, so a single odd reading cannot set off a false alarm. Then it turns the sensor's voltage into a UV index number.

Used by: runFlowchart, handleCommand, calibrateToKnownUv **Uses:** sortNumbers, debugLog

```
function phoneSendingOnlineUv
  set age to running time (ms) - onlineUvTime
  set maxAge to onlineUvMaxAgeMinutes * 60000
  return onlineUvTime > 0 and age <= maxAge
```

phoneSendingOnlineUv A yes-or-no question the program asks.

Asks: has the phone app sent an online UV index in the last 30 minutes?

Used by: chooseUvValue, showUvReading, sendStatus

```
function moderateOrHighUv
  if protectionLevel <= 0 then
    set stateName to 'alert'
    call showMessage 'Sunscreen time'
    call flashAndBeepUntilA false
    if ackPressed then
      call showMessage '1 tsp SPF 50+ on each arm, each leg, front, back...'
      call sunscreenApplied 1
    else if deviceOn then
      call standbyUntilButton
    else
      call alreadyProtected
```

moderateOrHighUv A step of the program.

UV 3 to 7: flashes and beeps slowly until A is pressed, then shows how much sunscreen to put on and starts the 2 hour timer. If sunscreen is already on, it just shows a tick and the time left. If nobody presses A within 10 minutes, the device assumes it is not being worn and goes to sleep.

Used by: runFlowchart **Uses:** sunscreenApplied, alreadyProtected, flashAndBeepUntilA, standbyUntilButton, showMessage

```
function wait5Minutes
  set stateName to 'wait'
  set ackPressed to false
  set waitEndTime to running time (ms) + wait5
  set timeLeft2 to wait5
  while deviceOn and timeLeft2 >= 0 and not call reapplyTimerExpired
    do
      call oledFooter join 'Next' call formatTime timeLeft2 call reapplyCountdownText
      if ackPressed then
        set ackPressed to false
        call showLedStatus
      +
      pause (ms) 1000
      set timeLeft2 to waitEndTime - running time (ms)
```

wait5Minutes A step of the program.

Waits 5 minutes before the next check. The wait ends early if the device is turned off or the sunscreen timer runs out. Pressing A during the wait shows the UV number and the countdown.

Used by: runFlowchart **Uses:** reapplyTimerExpired, showLedStatus, oledFooter, formatTime, reapplyCountdownText

```
function uvBand uv
  set rounded to round uv
  set band to 'extreme'
  if rounded <= uvLowMax then
    set band to 'low'
  else if rounded <= uvHighMax then
    set band to 'modhigh'
  else if rounded <= uvVeryHighMax then
    set band to 'vhigh'
  return band
```

uvBand Works out a piece of text - needs: uv.

Turns a UV index into one of the four levels on the flowchart: low (0 to 2), moderate or high (3 to 7), very high (8 to 10) or extreme (11 and above).

Used by: runFlowchart, sendStatus

```
function sensorReadingInRange
  set inRange to true
  if sensorRaw >= 1020 then
    set inRange to false
  +
  if sensorNoise >= 200 then
    set inRange to false
  +
  if sensorUv <= -1.5 or sensorUv >= 20 then
    set inRange to false
  +
  if inRange then
    set sensorErrorReported to false
  +
  return inRange
```

sensorReadingInRange A yes-or-no question the program asks.

Asks: is this reading believable? The answer is no if the sensor is missing or shorted, if the readings jump about (a loose wire), or if the number could not be real.

Used by: runFlowchart

```
function showCheckSensor
  set stateName to 'error'
  if not sensorErrorReported then
    call sendLine 'Sensor error:msg=CHECK SENSOR'
    set sensorErrorReported to true
  +
  show icon
  call showMessage 'CHECK SENSOR'
  call oledFooter join pin call formatNumber sensorVolts 'V'
  call beep 440 200
  call beep 440 200
  pause (ms) sensorRetrySeconds * 1000
```

showCheckSensor A step of the program.

When the reading is not believable: shows a cross and the words CHECK SENSOR, beeps twice, tells the app there is a problem, and tries again in 5 seconds. A faulty sensor is never treated as safe.

Used by: runFlowchart **Uses:** showMessage, oledFooter, beep, sendLine, formatNumber

```
function chooseUvValue
  set currentUv to max of 0 and sensorUv
  if call phoneSendingOnlineUv then
    set currentUv to max of currentUv and onlineUv
```

chooseUvValue A step of the program.

Picks the UV number the device will act on: the higher of the sensor's reading and the phone's online value, or just the sensor's reading when the phone is not sending.

Used by: runFlowchart **Uses:** phoneSendingOnlineUv

```
function showHappyFace
  set stateName to 'safe'
  show icon
  call oledShowMessage 'Low UV. No sunscreen needed'
```

showHappyFace A step of the program.

UV 0 to 2: shows a happy face. No sunscreen needed right now.

Used by: runFlowchart **Uses:** oledShowMessage

```
function veryHighUv
  if protectionLevel <= 2 then
    set stateName to 'alert'
    call showMessage 'Very high UV'
    call flashAndBeepUntilA true
    if ackPressed then
      call showMessage '1 tsp SPF 50+ on each arm, each leg, front, back...'
      call sunscreenApplied 2
    else if deviceOn then
      call standbyUntilButton
    +
    call alreadyProtected
```

veryHighUv A step of the program.

UV 8 to 10: flashes and beeps quickly until A is pressed, then shows the sunscreen advice plus hat and shade, and starts the 2 hour timer. It alerts again if the wearer so far only had the moderate advice.

Used by: runFlowchart **Uses:** sunscreenApplied, alreadyProtected, flashAndBeepUntilA, standbyUntilButton, showMessage

```
function extremeUv
  set stateName to 'alert'
  call tapArmUntilA
  if deviceOn then
    call showMessage 'Go inside now. Sunscreen on the way'
    if ackPressed then
      call sendEvent 'inside'
      call standbyUntilButton
    +
    call alreadyProtected
```

extremeUv A step of the program.

UV 11 or more: taps the arm until A is pressed (after 2 minutes it beeps instead), shows Go inside now, then sleeps until a button is pressed.

Used by: runFlowchart **Uses:** tapArmUntilA, standbyUntilButton, showMessage, sendEvent

```
function start2HourReapplyTimer
  set reapplyTimerRunning to true
  set reapplyDueTime to running time (ms) + reapply5
```

start2HourReapplyTimer A step of the program.

Starts the 2 hour countdown to the next sunscreen.

Used by: sunscreenApplied

```
function showReapplySunscreen
  set stateName to 'alert'
  set reapplyTimerRunning to false
  set protectionLevel to 0
  call sendEvent 'reapply'
  show icon
  call showMessage 'Reapply sunscreen'
```

showReapplySunscreen A step of the program.

Shows Reapply sunscreen and clears the timer, so the next trip round the flowchart alerts again if the sun is still strong.

Used by: runFlowchart **Uses:** showMessage, sendEvent

```
function deviceTurnsOn
  set stateName to 'on'
  show icon
  call oledHeader
  if demoMode then
    call oledShowMessage 'Device on (demo)'
  else
    call oledShowMessage 'Device on'
  +
  call oledFooter 'Waiting for phone'
  call servoOff
  call sendEvent 'on'
  pause (ms) 1500
  set deviceOn to true
```

deviceTurnsOn A step of the program.

Switching on: shows a tick, says Device on on the screen, tells the app, and marks the device as on.

Used by: on button A+B pressed, handleCommand, on start **Uses:** oledHeader, oledShowMessage, oledFooter, servoOff, sendEvent

3

Buttons

The micro:bit has two buttons. A means done: sunscreen is on, or I am going inside. B wakes the device up. Pressing both together turns it off, or on again.

3 blocks

```
on button A pressed
  set ackPressed to true
  set anyButtonPressed to true
  set skipMessage to true
```

on button A pressed Runs each time button A is pressed.

Pressing A means done: sunscreen is on, or I am going inside. It also wakes the device, skips a long message, and during a wait shows the UV number and the countdown.

```
on button B pressed
  set anyButtonPressed to true
```

on button B pressed Runs each time button B is pressed.

Pressing B wakes the device from sleep. Holding B while switching on gives the fast demo timings.

```
on button A+B pressed
  if deviceOn then
    call deviceTurnsOff
  else
    call deviceTurnsOn
```

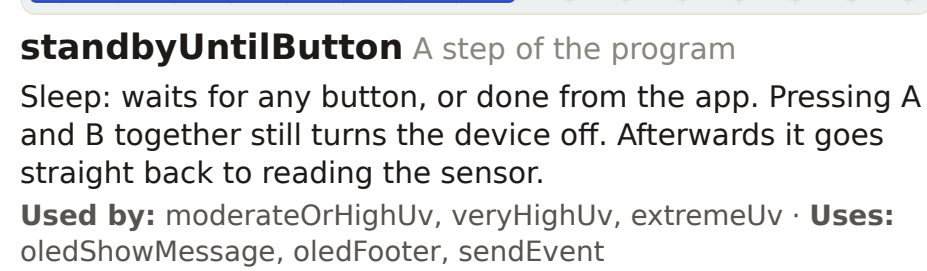
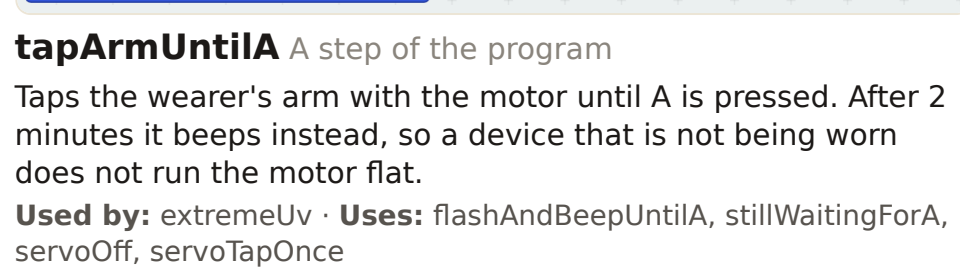
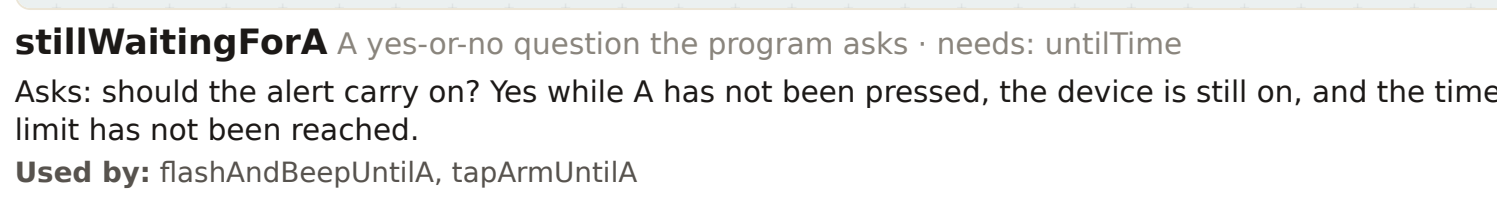
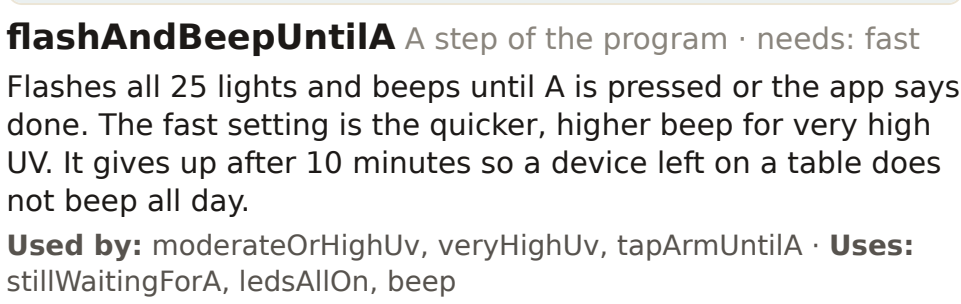
on button A+B pressed Runs when A and B are pressed together.

Pressing A and B together turns the device off, or on again.

Uses: deviceTurnsOn, deviceTurnsOff

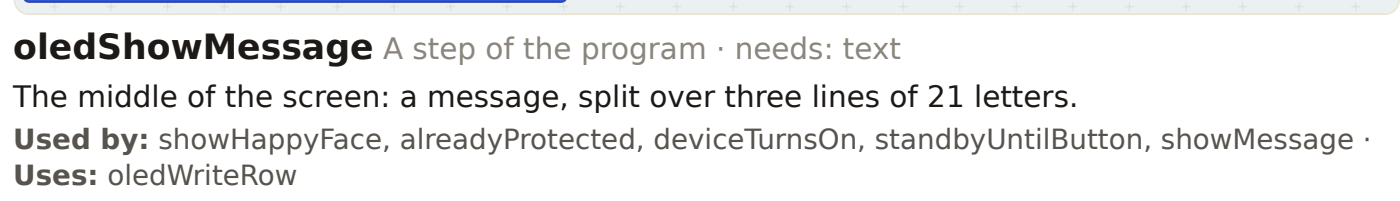
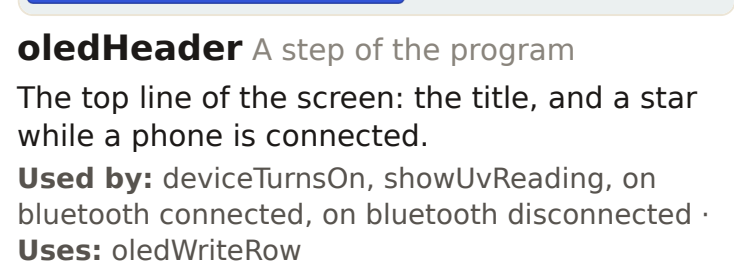
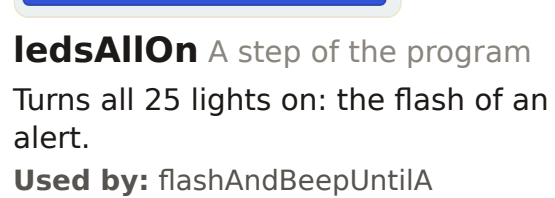
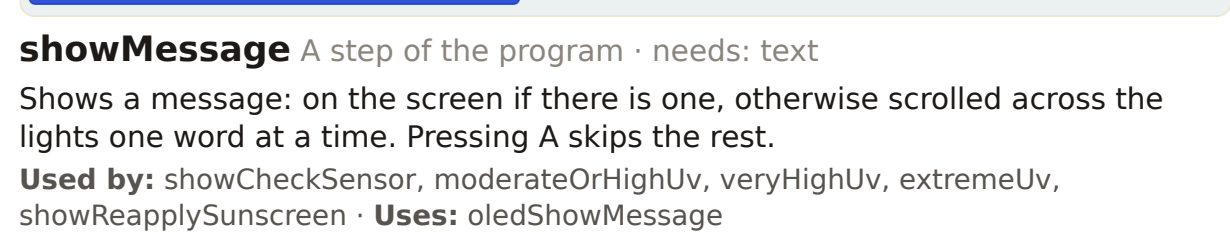
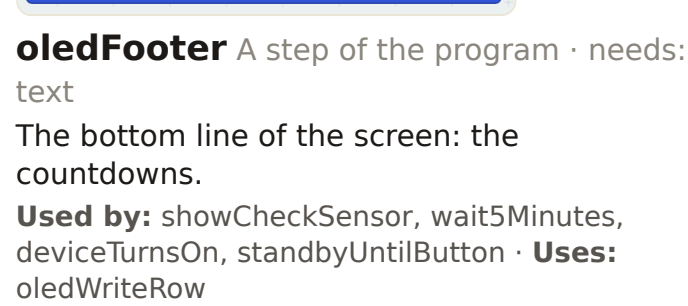
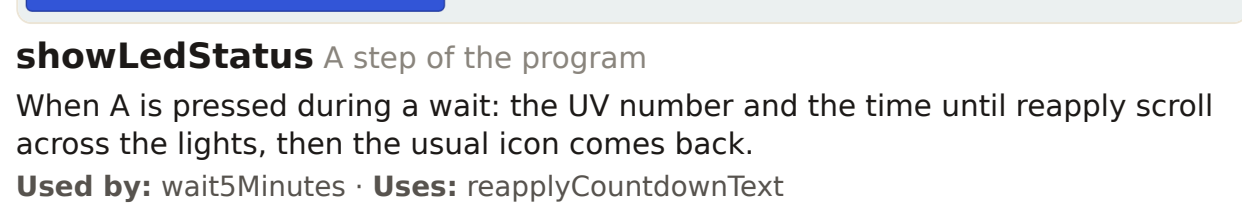
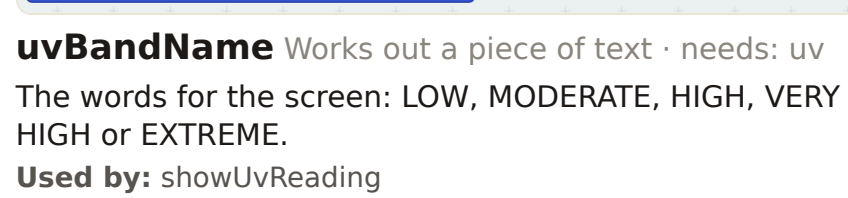
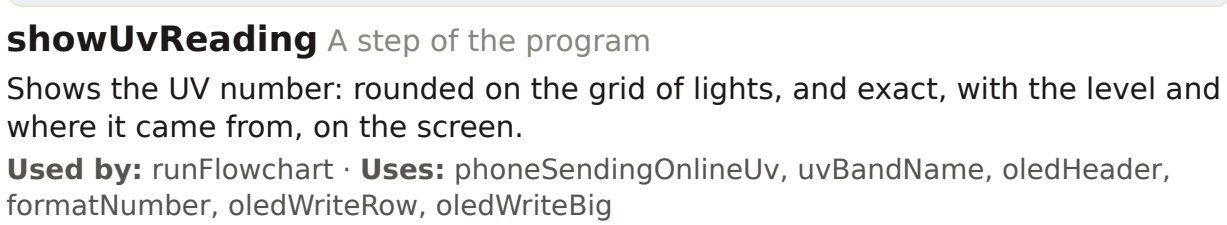
4 blocks

The ways the device gets your attention: flashing and beeping, tapping your arm with the small motor, and going to sleep when nobody answers. Each one keeps going until you press A or its time limit runs out. No answer for a long time means the device is probably not being worn, so it sleeps until a button is pressed.



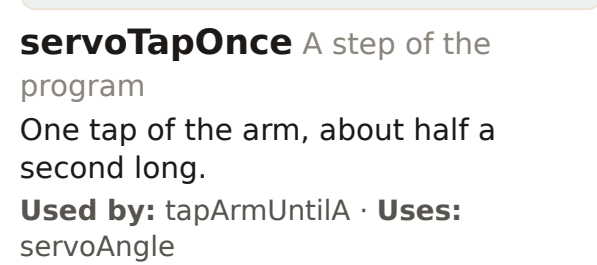
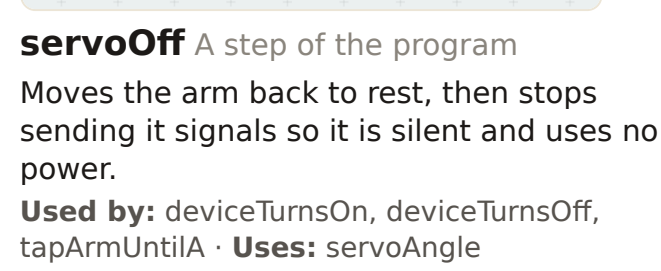
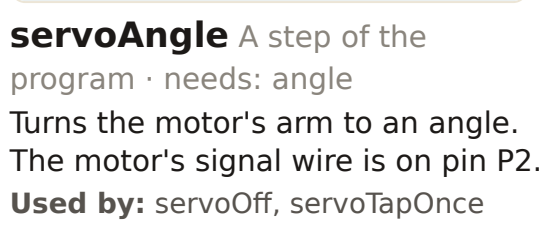
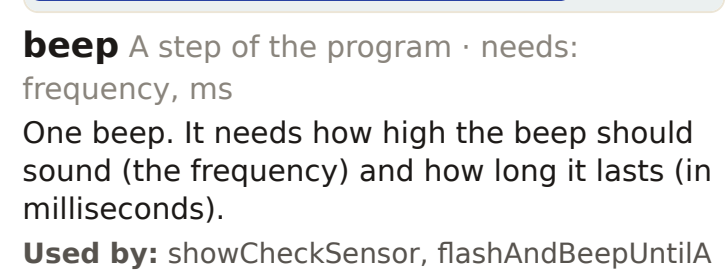
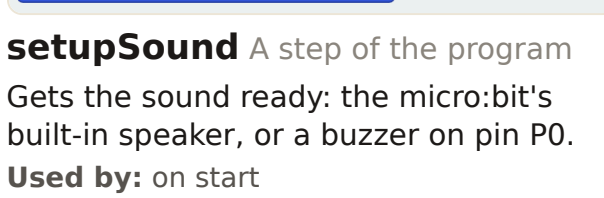
8 blocks

What you see. The 5 by 5 grid of lights shows the UV number, a happy face or a warning, and messages one word at a time. If a small screen is fitted, it shows the number large, the level, the message and the countdowns.



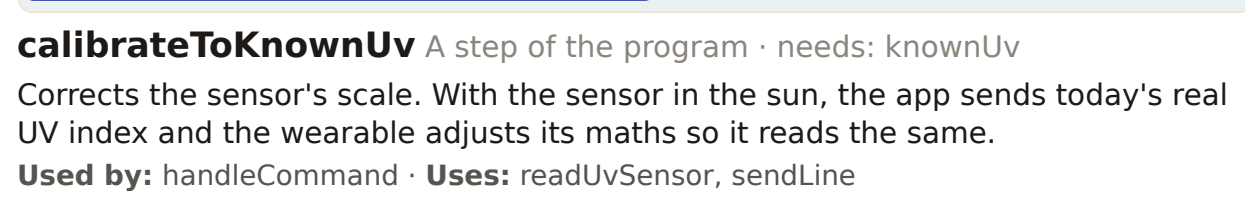
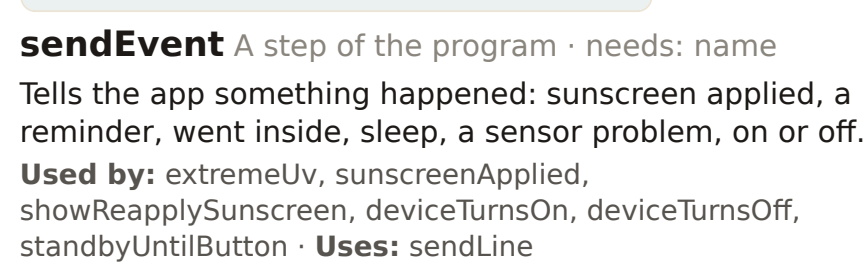
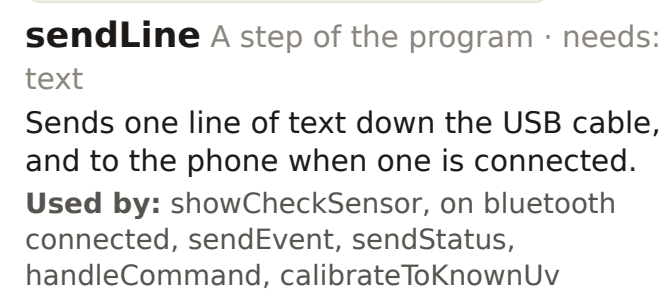
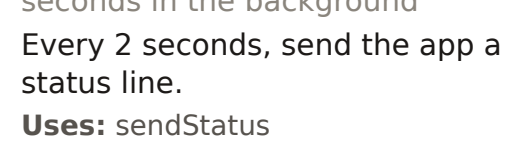
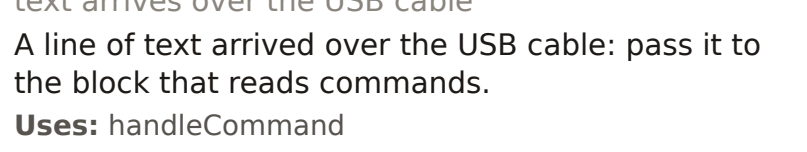
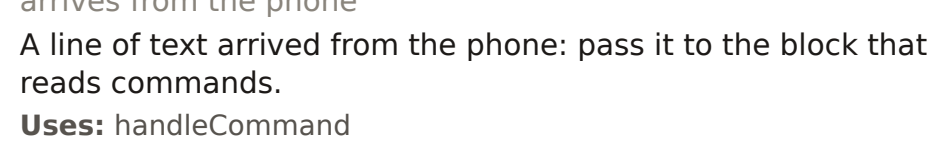
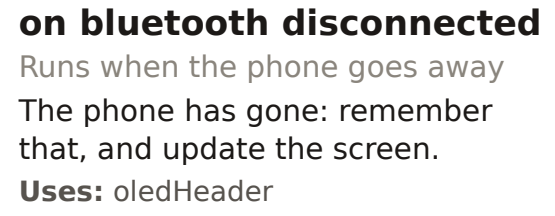
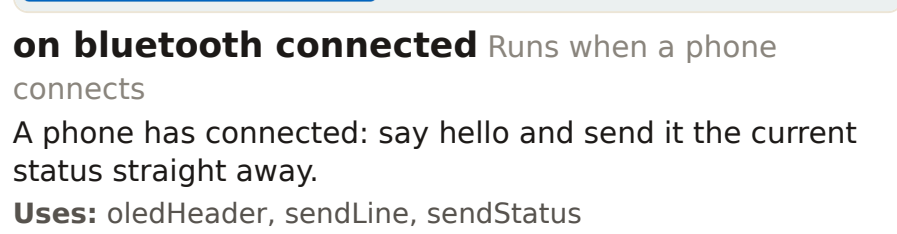
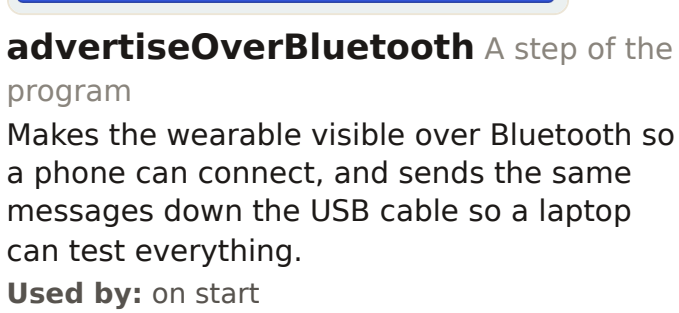
5 blocks

The beeps, and the small motor (a servo) whose arm taps your wrist when the sun is extreme.



1 blocks

How the wearable and the phone app talk over Bluetooth. The phone sends the online UV index for where you are; the wearable sends back what it is doing every two seconds and reports each event, such as sunscreen applied. The same messages also go down the USB cable, so everything can be tested on a laptop.




```
function oledInitCommands
call oledCommand 213
call oledCommand 128
call oledCommand 168
call oledCommand 63
call oledCommand 211
call oledCommand 8
call oledCommand 64
if <oledType> == <2> then
call oledCommand 173
call oledCommand 139
else
call oledCommand 141
call oledCommand 28
call oledCommand 32
call oledCommand 2
call oledCommand 161
call oledCommand 260
call oledCommand 218
call oledCommand 18
call oledCommand 129
call oledCommand 207
call oledCommand 217
call oledCommand 241
call oledCommand 219
call oledCommand 64
call oledCommand 164
call oledCommand 166
call oledCommand 175
```

oledInitCommands A step of the program
Blanks the whole screen.
Used by: deviceTurnsOff, setupOled - Uses: oledCommand

```
function oledClear
if <oledPresent> then
for <clearPage> from 0 to <7>
do
call oledSetPosition <clearPage> <0>
repeat <45> times
do
call oledData <0> <0> <0>
```

oledClear A step of the program
Blanks the whole screen.
Used by: deviceTurnsOff, setupOled - Uses: oledData, oledSetPosition

```
function oledWriteText row col text
if <oledPresent> then
set <smallText> to <text>
if <length of smallText> > <21> <col> then
set <smallText> to substring of <smallText> from <0> of length <21> <col>
call oledSetPosition row col <0>
for <charIndex> from 0 to <length of smallText> <col>
do
set <charCode> to char code from <smallText> at <charIndex>
call oledData call fontByte <charCode> <0> call fontByte <charCode> <1> call fontByte <charCode> <2>
call oledData call fontByte <charCode> <3> call fontByte <charCode> <4> <0>
```

oledWriteText A step of the program - needs: row, col, text
Writes small text: up to 21 characters on each of 8 rows.
Used by: oledWriteRow - Uses: fontByte, oledData, oledSetPosition

```
function oledWriteRow row text
set rowText to <text>
while <length of rowText> <= <21>
do
set rowText to join rowText < >
call oledWriteText row <0> rowText
```

oledWriteRow A step of the program - needs: row, text
Writes a whole row, filling the rest with spaces so old text is wiped.
Used by: deviceTurnsOff, showUvReading, oledHeader, oledShowMessage, oledFooter - Uses: oledWriteText

```
function oledWriteBig row col text
if <oledPresent> then
set <bigText> to <text>
if <length of bigText> > <18> then
set <bigText> to substring of <bigText> from <0> of length <18>
call oledSetPosition row <0> <col> <0>
for <half> from 0 to <1>
do
call oledSetPosition row <+> <half> <col> <+> <0>
for <bigCharIndex> from 0 to <length of bigText> <col>
do
set <bigCharCode> to char code from <bigText> at <bigCharIndex>
for <columnIndex> from 0 to <5>
do
if <columnIndex> <= <5> then
set <glyphColumn> to call fontByte <bigCharCode> <columnIndex>
if <half> == <0> then
<stretchedColumns> add value <nibbleStretch> get value at remainder of <glyphColumn> / <16> to end
else
<stretchedColumns> add value <nibbleStretch> get value at floor * <glyphColumn> / <16> to end
call oledData <stretchedColumns> get value at <0> <stretchedColumns> get value at <0> <stretchedColumns> get value at <1>
call oledData <stretchedColumns> get value at <1> <stretchedColumns> get value at <2> <stretchedColumns> get value at <2>
call oledData <stretchedColumns> get value at <3> <stretchedColumns> get value at <3> <stretchedColumns> get value at <4>
call oledData <stretchedColumns> get value at <4> <stretchedColumns> get value at <5> <stretchedColumns> get value at <5>
```

oledWriteBig A step of the program - needs: row, col, text
Writes double-size text, two rows tall and up to 10 characters. It is used for the big UV number.
Used by: showUvReading - Uses: fontByte, oledData, oledSetPosition